

Robust Multi-View Pig Posture Recognition in Commercial Farming Environments

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Abstract—Monitoring pig posture provides valuable indicators of animal health and welfare, but manual observation is not scalable in commercial farming environments. Computer vision offers a promising solution; however, real-world deployment is challenged by significant distribution shifts, including variations in camera viewpoints, lighting conditions, animal appearance, temporal redundancy, and severe class imbalance. These factors often lead to models overfitting to spurious, camera-specific features and failing to generalize to unseen conditions. In this work, we propose a unified deep learning pipeline for robust pig posture recognition under such real-world constraints. We frame the problem as viewpoint-level domain adaptation and employ a DINOv3 Vision Transformer backbone augmented with camera-type embeddings to incorporate domain context. To improve generalization, we combine multiple strategies: Deep CORAL for feature-level distribution alignment between source and target domains, a temporal reweighting scheme based on an exponential kernel to mitigate redundancy, and focal loss with inverse-frequency weighting and label smoothing to address class imbalance. Additional regularization is introduced via Cut-Mix augmentation and Sharpness-Aware Minimization (SAM). Model robustness is further enhanced through cross-camera validation and ensemble inference across multiple holdout splits. Experimental results demonstrate strong generalization to unseen camera viewpoints, achieving a cross-validation macro F1 score of 0.767 and 0.754 on the held-out test set, with consistent performance across distribution shifts. The proposed approach shows improved robustness to rare classes and ambiguous postures, while maintaining well-calibrated prediction confidence. These findings highlight the importance of jointly addressing domain shift, data redundancy, and class imbalance, and provide a practical framework for deploying scalable, vision-based livestock monitoring systems.

Index Terms—Pig posture recognition, Computer vision, Domain adaptation, Distribution shift, Precision livestock farming, Vision Transformers (ViT), Class imbalance, Temporal redundancy

I. INTRODUCTION

Pig posture monitoring in commercial farming environments provides actionable signals for animal health and welfare. For instance, lateral lying may indicate either rest or illness depending on its duration; transitions between standing and sitting reflect activity patterns; and sternal lying is often associated with thermal comfort. While these signals are valuable, manual observation is inherently unscalable, making computer vision a natural solution. However, real-world deployment

conditions differ substantially from the controlled settings in which most posture recognition models are developed.

Commercial pig barns present a combination of visual, geometric, and data-related challenges that hinder generalization. Scenes are cluttered, with frequent inter-animal occlusions and highly variable lighting, while diverse fixed camera viewpoints introduce shifts in perspective geometry, lens distortion, and background structure. Models trained on limited viewpoints therefore tend to overfit to camera-specific cues (e.g. pen layouts or feeder positions) instead of learning posture-relevant features, leading to failures on unseen views. This issue is compounded by the morphology of livestock pigs: their rounded bodies and short limbs reduce visible articulation cues, making posture distinctions inherently difficult, even to humans, especially under top-down or oblique views. At the dataset level, temporal redundancy further amplifies overfitting, as densely sampled frames (≈ 5 -second intervals) produce near-duplicate training examples; meanwhile, strong class imbalance dominated by lying postures introduces biases in learning and disproportionately penalizes performance under macro-averaged F1.

This problem is formalized in the Multi-view Pig Posture Recognition challenge [1], where models are trained on images from five camera viewpoints and evaluated on three disjoint, unseen viewpoints. The challenge is further complicated by a distribution shift in the animals themselves: test data comes from a different production batch, with variations in breed, size, and coat appearance, preventing models from relying on memorization of individual pigs.

To address these challenges, we propose a unified pipeline targeting viewpoint shift, temporal redundancy, and class imbalance. We frame the problem as domain adaptation at the level of camera viewpoint, aiming to learn representations that are both discriminative for posture and invariant to viewpoint. To bridge the distribution gap between labeled source views and unlabeled target views, we employ explicit covariance alignment via Deep CORAL. Temporal redundancy is mitigated through a density-based reweighting scheme using an exponential (Ornstein–Uhlenbeck) kernel, which downweights near-duplicate frames and promotes temporally diverse sampling during training. Class imbalance is handled using focal loss with inverse-frequency class weighting and label

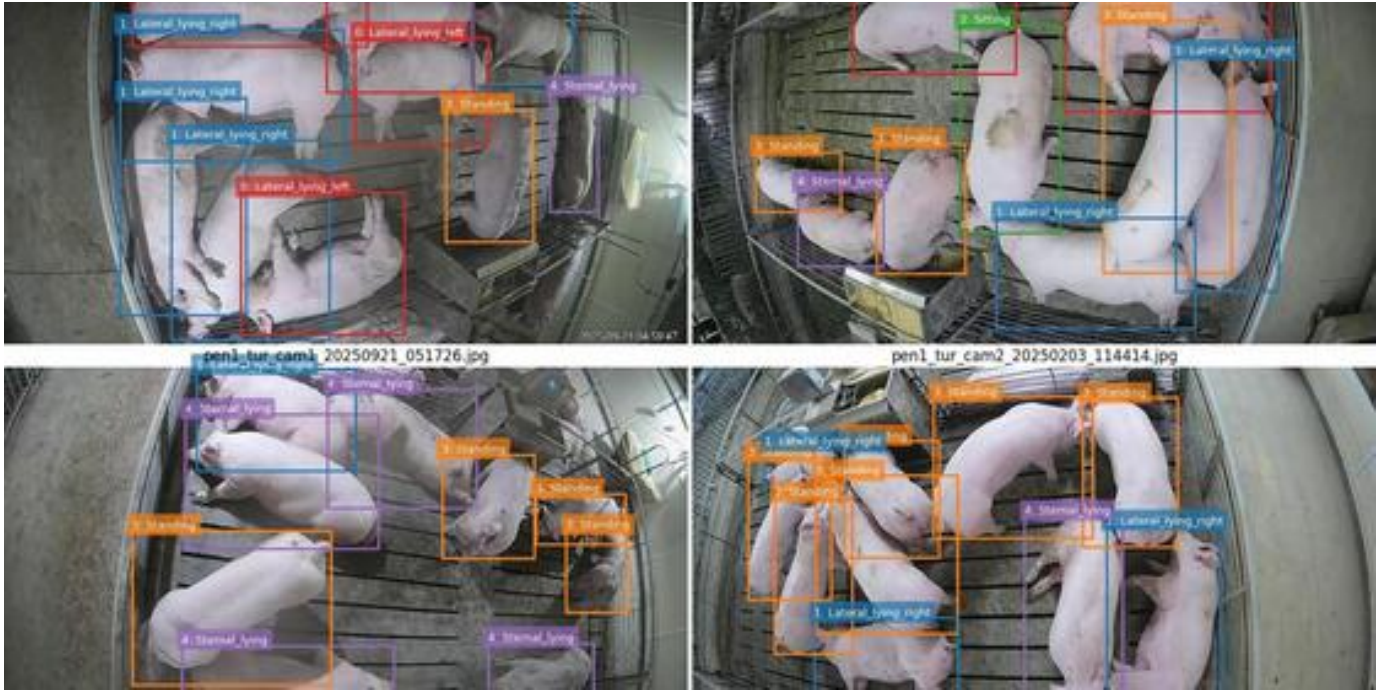


Fig. 1. Example of commercial pig barn pictures with pigs in bounding boxes

smoothing, improving sensitivity to rare postures. Additional regularization is introduced through CutMix augmentation and Sharpness-Aware Minimization (SAM), encouraging convergence to flatter minima that generalizes better under distribution shift. We further incorporate a learned camera-type embedding to explicitly condition the model on observation context (e.g., orb versus tur environments). Finally, we apply a prediction ensemble across six cross-validation folds, each defined by a distinct orbxtur holdout pair, enabling models with complementary viewpoint exposure to collectively mitigate individual blind spots.

II. BACKGROUND AND RELATED WORK

A. Pig Posture and Behaviour Recognition

Early work in automated pig behaviour monitoring relied on hand-crafted features (e.g. background subtraction, ellipse fitting, and motion energy images) applied to top-down camera views [2]. Deep learning subsequently improved accuracy, with CNN-based classifiers (ResNet, EfficientNet) applied to cropped pig bounding boxes achieving strong performance under fixed-camera conditions [3], [4]. However, these methods are typically not evaluated on unseen camera viewpoints, limiting their robustness in multi-camera deployments. Moreover, approaches that incorporate pose estimation often simplify the task to coarse binary distinctions (e.g., lying vs. not lying), whereas this work considers a finer-grained taxonomy including multiple lying postures, as well as sitting and standing.

B. Domain Adaptation for Visual Recognition

Domain adaptation addresses the problem of training on a source distribution and deploying on a shifted target distribu-

tion. Two families of methods are relevant:

1) *Statistical moment matching*: CORAL [5] aligns second-order statistics (covariance matrices) between source and target feature distributions. Deep CORAL [5] integrates this as a differentiable loss. Maximum Mean Discrepancy (MMD) [6] and its variants align higher-order moments. These methods complement adversarial approaches by providing explicit distributional alignment.

C. Foundation Models and Transfer Learning

Vision Transformers [7] pre-trained with self-supervised objectives like DINO [8] and DINOv2 [13] produce features that transfer strongly to downstream tasks with limited labelled data. DINOv3 [13] further refines this approach. These models are particularly valuable in our setting, where training data is limited ($\approx 23k$ annotated instances across $\approx 3k$ images) and the domain shift requires strong general-purpose representations that can be selectively adapted.

D. Sharpness-Aware Minimisation

Foret et al. [15] proposed SAM, which seeks model parameters that lie in flat regions of the loss landscape rather than sharp minima. Flat minima generalize better because small perturbations to the input distribution (such as a change in camera viewpoint) cause smaller changes in loss. SAM has been shown to improve generalization in low-data regimes and under distribution shift [9].

E. Focal Loss and Class Imbalance

Lin et al. [18] introduced Focal Loss to address the extreme foreground-background imbalance in object detection. The

modulating factor $(1 - p_t)^\gamma$ down-weights easy examples and focuses learning on hard, misclassified instances. Combined with per-class inverse-frequency weighting and label smoothing [10], it is well-suited for imbalanced multi-class classification with macro-averaged metrics.

III. METHODS

A. Research Objectives

This work focuses on developing a robust computer vision system for pig posture recognition that can operate reliably in real-world commercial farming environments. In practice, such environments introduce significant variability, including differences in camera viewpoints, lighting conditions, and animal appearance. These variations often degrade model performance and limit the scalability of vision-based welfare monitoring systems. To address this, the project aims to learn representations that remain discriminative for posture classification while being invariant to nuisance factors such as viewpoint and environmental artefacts. By improving robustness to these sources of variation, this work seeks to reduce a key barrier to deploying automated, large-scale livestock monitoring systems.

O1: Model Development for Robust Posture Recognition.

To design and train a computer vision model that accurately classifies pig postures while generalizing across unseen camera viewpoints and variations in animal appearance. This involves developing representations that prioritize posture-relevant features and suppress spurious correlations introduced by viewpoint, background, or lighting differences.

O2: Evaluation Under Distribution Shift. To systematically evaluate the model’s generalization performance under realistic distribution shifts, including unseen camera angles, cross-batch variation in pigs, temporal redundancy, and class imbalance. This objective aims not only to quantify performance but also to analyze how different sources of distribution shift affect model reliability, providing insights to guide the design of more robust and transferable precision livestock farming systems.

B. Research Methodology

1) *Data Acquisition:* The dataset was provided by the Smart Sensing Lab at Michigan State University as part of the Multi-view Pig Posture Recognition competition. Data were collected at the Swine Research Center, Michigan State University, from two pig pens instrumented with eight overhead cameras, as shown in Figure 2. Three pig breeds are represented: Purebred Yorkshire, Exotic crossbred, and Meishan.

2) *Data Content:* Each instance in the dataset includes:

- An axis-aligned bounding box $(x_{\min}, y_{\min}, w, h)$,
- A posture label (training only),
- The name of the camera, which indicates its type: turret “tur” or orbital “orb”,
- A timestamp in the format YYYYMMDD_HHMMSS (from the filename).

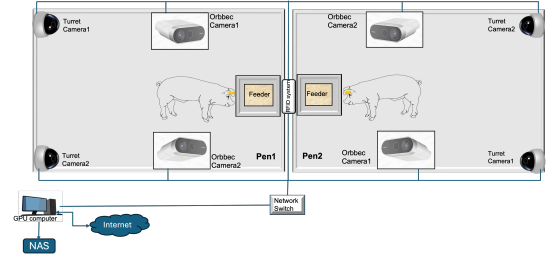


Fig. 2. Camera array used in data collection.

The five posture classes are: Lateral lying left (0), Lateral lying right (1), Sitting (2), Standing (3), and Sternal lying (4), as illustrated in Figure 3.

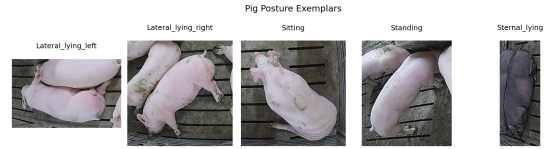


Fig. 3. Example pig postures with corresponding class labels.

3) *Data Split and Distribution:* Table I summarizes the composition of training and test sets, including the number of images, instances, camera viewpoints, and viewpoint names.

TABLE I
DATASET COMPOSITION ACROSS TRAINING AND TEST SETS.

Property	Training Set 1 (T1)	Training Set 2 (T2)	Test Set
Images	3,090	3,150 (+60)	~1,300
Instances	22,934	23,450 (+516)	11,708
Camera viewpoints	5	5 + 3 (bridge)	3
Viewpoint names	pen1_orb_cam1, pen1_orb_cam2, pen1_tur_cam2, pen2_orb_cam1, pen2_tur_cam1	T1 viewpoints + pen1_tur_cam1, pen2_orb_cam2, pen2_tur_cam2	pen1_tur_cam1, pen2_orb_cam2, pen2_tur_cam2

Table II summarizes the distribution of camera types and posture labels across the dataset, while Table III reports inter-frame time gap statistics for the orb and tur cameras. These statistics highlight both class imbalance and temporal redundancy, which are important considerations for sampling and augmentation strategies during model training.

TABLE II
CAMERA TYPE AND POSTURE DISTRIBUTION.

Camera Type	Instances	(%)
orb	7,952	22.6
tur	27,206	77.4

Posture Class	Count	(%)
Lateral_lying_left	3,083	13.1
Lateral_lying_right	3,435	14.6
Sitting	695	3.0
Standing	9,928	42.3
Sternal_lying	6,309	26.9

TABLE III
INTER-FRAME TIME GAP STATISTICS FOR ORB AND TUR CAMERAS.

Statistic	orb	tur
Min	1 s	7 s
25th percentile	5 s	90 s
Median	5 s	120 s
75th percentile	60 s	200 s
Max	18,739,940 s	21,028,708 s
Mean	13,484 s	15,485 s
Std	498,720 s	521,758 s

4) *Preprocessing and Augmentation*: Each pig instance is first localized using the provided bounding box annotations and cropped from the full image, with an additional 10% padding applied on all sides to preserve surrounding contextual cues. The resulting crops are resized to 256×256 pixels to match the native input resolution of the DINOv3 ViT-B/16 backbone, and subsequently normalized using ImageNet statistics.

During training, we apply a set of stochastic augmentations to improve robustness and invariance. In addition to visual inputs, we incorporate metadata to capture domain-specific variation: the camera view is used to define holdout validation splits, and camera type (orbital or turret) is encoded as a learned 32-dimensional embedding, which is concatenated with the backbone feature representation to provide an explicit domain-aware input to the classifier. The specific augmentation operations and parameters are summarized below.

TABLE IV
DATA AUGMENTATION STRATEGIES AND PARAMETERS USED DURING TRAINING.

Augmentation	Parameters / Description
Random horizontal flip	$p = 0.5$, with class swap ($0 \leftrightarrow 1$)
Random rotation	$\pm 15^\circ$
Colour jitter	brightness 0.3, contrast 0.3, saturation 0.2, hue 0.05
Random erasing	$p = 0.2$, scale 2–15%
CutMix	$\alpha = 1.0$, $p = 0.5$; mixes images and labels proportionally to cut area

C. Sample Reweighting

The training data exhibits high temporal redundancy, which can lead to overfitting. To mitigate this, we implement a per-camera exponential temporal similarity kernel, inspired by the Ornstein–Uhlenbeck process [11]. For each image timestamp t_i from camera c , we compute a local density:

$$d_i = \sum_{\substack{j \in \text{camera}(c) \\ j \neq i}} \exp\left(-\frac{|t_i - t_j|}{\tau}\right), \quad w_i = \frac{1}{1 + d_i},$$

so that frames with many temporally adjacent neighbors are downweighted, while isolated frames receive weights close to 1. All bounding-box crops from the same image inherit its weight. The computation is truncated to a sliding window of $\pm 5\tau$ per camera, yielding $O(N \log N)$ complexity.

We derive τ empirically from the data rather than treating it as a hyperparameter. We define a pairwise redundancy function

$$R(\Delta t) = E[\text{IoU}(i, j) \cdot \mathbf{1}[y_i = y_j] \mid |t_i - t_j| = \Delta t, \text{ same camera}],$$

which captures spatial and postural overlap over time. After subtracting the asymptotic baseline $R_\infty = 0.491$, we normalize to obtain

$$\tilde{R}(\Delta t) = \frac{R(\Delta t) - R_\infty}{R(0) - R_\infty}, \quad R(0) = 0.836.$$

Assuming $\tilde{R}(\Delta t) \approx \exp(-\Delta t/\tau)$, we estimate τ via weighted log-linear regression, giving

$$\tau = 78.9 \text{ s}.$$

Under this weighting, the dataset’s 3,150 images correspond to 928 effective independent images (29.4%), with dense orb cameras showing extreme redundancy and sparser tur cameras retaining most effective samples. The temporal weights are applied multiplicatively with bridge-domain oversampling and used in a weighted random sampler. During training, we sample with replacement according to these weights for each epoch, ensuring that high-weight, temporally isolated frames are more likely to appear while dense bursts are downsampled. This approach preserves the benefits of the exponential temporal kernel while stabilizing training and improving coverage of rare postures.

D. Model Architecture and Training Procedure

During the initial training stage, we freeze the classification head and update only the backbone. We adopt this strategy following prior work [12], which showed that staged training helps preserve pretrained representations and improves stability by reducing catastrophic forgetting. By freezing the head, we encourage controlled adaptation of the shared feature space before jointly optimizing task-specific components, leading to more stable convergence and improved generalization.

1) *Phase A – Frozen Backbone*: The model comprises a shared DINOv3 ViT-Base/16 backbone and a posture classification head, connected via a fused feature representation:

$$\mathbf{f} = [\text{backbone}(\mathbf{x}) \parallel \text{Embed}_{\text{cam}}(c)] \in R^{800}$$

where $\mathbf{x}$ denotes the input image and $c \in \{\text{orb}, \text{tur}\}$ represents the camera type. The 768-dimensional backbone feature is concatenated with a learned 32-dimensional camera-type embedding, producing an 800-dimensional joint representation that encodes both visual and domain-specific information.

The posture classification head consists of two fully connected layers ($800 \rightarrow 512 \rightarrow 5$), interleaved with Batch Normalization and ReLU activations, producing logits over the five posture classes.

During Phase A, the DINOv3 backbone is frozen. Only the classification head and the camera-type embedding are trained using AdamW with learning rate 10^{-3} and weight

decay 10^{-4} [14]. Phase A lasts for 7 epochs. Freezing the backbone prevents randomly initialized head parameters from corrupting the pre-trained representations.

2) *Phase B – Full Fine-Tuning*: In Phase B, the backbone is unfrozen and all model parameters are optimized. We use Sharpness-Aware Minimization (SAM) [15] wrapped around AdamW with $\rho = 0.05$, initial learning rate 10^{-5} , and cosine annealing to 10^{-7} . Phase B lasts for 28 epochs, allowing the entire network to adapt to the posture recognition task while preserving the pre-trained backbone features.

E. Loss Function

We optimize the model using a weighted combination of two losses:

$$\mathcal{L} = \mathcal{L} * \text{focal} + \lambda_c \mathcal{L} * \text{CORAL}$$

with $\lambda_c = 0.1$.

The focal loss is computed with label smoothing ($\epsilon = 0.1$) and per-class inverse-frequency weighting α_i :

$$\mathcal{L}_{\text{focal}} = - \sum_i \alpha_i (1 - p_i)^\gamma \log(p_i), \quad \gamma = 2.0$$

As an auxiliary loss function, we incorporate CORAL to align feature distributions between datasets. At each training step, we forward a batch of unlabelled test images through the shared backbone to obtain target features $\mathbf{f}$ (detached from the computational graph). The CORAL loss minimizes the difference in covariance matrices between the training (source) and test (target) features:

$$\mathcal{L} * \text{CORAL} = \frac{1}{4d} \|C * \text{source} - C_{\text{target}}\|_F^2$$

where $d = 800$ is the dimensionality of the fused feature vector. During Phase A (frozen backbone), CORAL gradients only update the classification head and camera embedding; in Phase B (unfrozen backbone), CORAL provides meaningful gradients to reshape the backbone features toward the test distribution. When target features are unavailable, we fall back to random domain-pair alignment within the training data.

We adopt a $\frac{1}{4d}$ normalization rather than the original $\frac{1}{4d^2}$ used in Deep CORAL. With $d = 800$, the d^2 normalization produces a denominator of 2,560,000, rendering the loss negligible compared to the focal loss. Our $\frac{1}{4d}$ normalization (denominator 3,200) ensures that CORAL contributes a meaningful gradient signal alongside the focal loss.

F. Validation

We adopt a cross-camera holdout scheme that withholds one `orb` and one `tur` domain simultaneously and uses them for validation. With 3 `orb` and 2 `tur` viewpoints in T1, this yields $3 \times 2 = 6$ holdout pairs, each producing a separate model.

For T2, bridge domains (the three test camera viewpoints) are always retained in training and excluded from the holdout pool; pairs are formed from the remaining domains, again yielding 6 combinations.

G. Inference

The six models obtained from distinct holdout pairs are combined via softmax probability averaging. All models share identical architecture and training hyperparameters (including random seed 42), differing only in the specific (`orb`, `tur`) domains held out during validation. At inference, each model produces softmax probabilities for every test instance, and the final ensemble prediction is given by the $\arg \max$ of the mean probability vector across all models. We further apply test-time augmentation (TTA) using horizontal flipping. Each test crop is evaluated in both its original and horizontally flipped orientations, yielding probability vectors $\mathbf{p}$ and $\mathbf{p}'$, respectively, with class `Later_Lying_Left` and `Later_Lying_Right` swapped in $\mathbf{p}'$ to account for left–right symmetry. The final prediction for each model is computed as

$$\arg \max \left(\frac{\mathbf{p} + \mathbf{p}'}{2} \right),$$

and these averaged probabilities are subsequently aggregated across all ensemble members.

H. Evaluation Metric

Performance is measured by macro-averaged F1 score:

$$F_1^{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{2 \cdot P_k \cdot R_k}{P_k + R_k}$$

where $K = 5$ classes and P_k , R_k are the precision and recall for class k . This gives equal weight to all classes regardless of frequency, penalizing models that sacrifice rare-class performance for overall accuracy.

IV. RESULTS

Our approach achieves a 6-fold cross-camera validation macro-averaged F1 score of 0.767, demonstrating strong generalization across unseen camera viewpoints. On the actual test set, the model attains a macro-averaged F1 score of 0.754 on the public 30% split, corresponding to approximately the 85th percentile on the Kaggle leaderboard of 162 participants (as of April 6, 2026). The close agreement between cross-validation and leaderboard performance indicates that the proposed validation scheme provides a reliable estimate of out-of-domain performance.

Class-wise analysis shows strong performance on visually distinct postures such as standing, achieving a 97.3% accuracy on par with prior literature, as shown in figure 4. The primary source of error arises from sternal lying, which is frequently confused with other lying postures due to its intermediate visual characteristics and sensitivity to viewpoint and occlusion.

During development, we observed that pseudo-labeling consistently degraded performance by approximately 2–3%, likely due to error propagation under domain shift. In contrast, label smoothing in the focal loss proved important: removing smoothing reduced performance, while its inclusion improved macroaveraged F1 by approximately 5–6%, indicating improved robustness to label noise and class ambiguity.

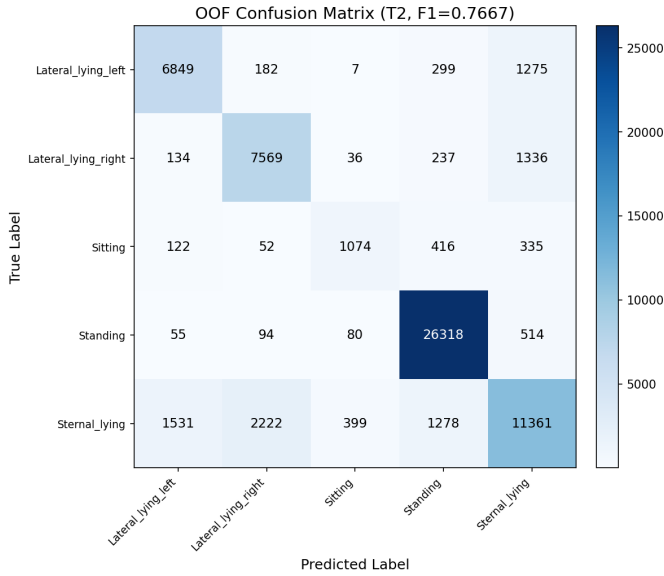


Fig. 4. Out of Fold Confusion Matrix

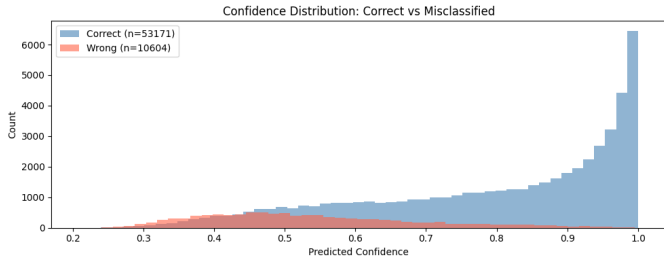


Fig. 5. Prediction Confidence

Despite using a ViT-B/16 backbone, the approach is not compute-saturated. Scaling to larger DINOv3 variants (e.g., ViT-L, ViT-H, or ViT-7B) would likely yield further gains, particularly given the strong performance of the current architecture and training pipeline. This suggests that the method is competitive and could approach top-tier leaderboard performance under higher compute budgets.

Finally, the model exhibits well-calibrated confidence estimates. Predictions are typically high-confidence when correct, while lower-confidence outputs correspond to genuinely ambiguous cases (see Figure 5). Notably, inspection of high-confidence misclassifications revealed instances of annotation error, where the model’s prediction was visually more consistent with the image than the annotated label. This suggests that part of the residual error may be attributable to annotation error rather than model deficiency.

V. CONCLUSION

Under the available computational constraints, the proposed approach does not yet enable fully automated posture recognition at deployment quality. However, the model’s strong performance and well-calibrated confidence estimates suggest it can substantially reduce human annotation burden by reliably

flagging high-confidence predictions and prioritizing uncertain cases for manual review.

The results further indicate that the method is not capacity-limited by design but by compute. Scaling to larger backbones (e.g., higher-capacity DINOv3 variants) and training on higher-quality annotations would likely yield significant performance improvements, potentially enabling fully automated systems in practical settings.

A. Limitations

This study is constrained by the use of a ViT-B/16 backbone, limited hyperparameter exploration, and potential label noise in the training data. Additionally, while the cross-camera validation protocol approximates real-world generalization, it may not fully capture all sources of domain shift encountered in deployment.

B. Future work

Several directions are promising for further improvement. First, scaling to larger backbone architectures is expected to improve representation quality. Second, a more systematic exploration of the hyperparameter space may yield additional gains. Third, expanding the ensemble to include models trained with diverse hyperparameters or training regimes could improve robustness. Fourth, alternative formulations such as weighted label smoothing may further enhance performance under class imbalance and annotation noise. Finally, augmenting the position of the bounding box relative to the full image as an additional feature should be explored, as it may provide positional and perspective context useful for distinguishing postures.

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